RP2040-LCD-1.28

Overview

RP2040-LCD-1.28 is a low-cost, high-performance MCU board designed by Waveshare. Although it is tiny, it incorporates a 1.28inch LCD round display, Li-ion battery charger, 6-axis sensor (3-axis accelerometer and 3-axis gyroscope) and so on, adapting all GPIO and Debug headers, which makes it easy for you to develop and integrate it into products quickly.

Feature

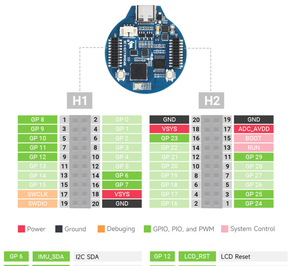
- RP2040 MCU chip designed by Raspberry Pi in the United Kingdom.
- Dual-core Arm Cortex M0+ processor, flexible clock running up to 133 MHz.
- 264KB of SRAM, and 2MB of onboard Flash memory.
- Type-C connector, keeps it up to date, easier to use.
- Onboard 1.28-inch 240 x 240 resolution, 65K RGB IPS LCD display for clear color pictures.
- Lithium battery recharge/discharge header, suitable for mobile devices.
- All GPIOs are adapted through 1.27 pitch female headers (There are 30 pins in total, but some pins have been connected to the internal circuit, you need to pay attention when multiplexing, please refer to the wiki for details).
- USB 1.1 with device and host support.
- Low-power sleep and dormant modes.
- Drag-and-drop programming using mass storage over USB.
- 2 x SPI, 2 x I2C, 2 x UART, 4 x 12-bit ADC, 16 × controllable PWM channels.
- Accurate clock and timer on-chip.
- Temperature sensor.
- Accelerated floating-point libraries on-chip.
- 8 x Programmable I/O (PIO) state machines for custom peripheral support.

Specification

LCD Parameter			
Controller	GC9A01A	Resolution	240 (H) RGB x 240(V)
Communication interface	SPI	Display Size	Φ32.4mm
Display Panel	IPS	Pixel Size	0.135 (H) x 0.135 (V) mm

IMU Parameter	
Sensor	QMI8658C
Accelerometer	Resolution: 16 bits Measurement Range (optional): ±2, ±4, ±8, ±16g
Gyroscope	Resolution: 16 bits Measurement Range (optional): ±16, ±32, ±64, ±128, ±256, ±512, ±1024, ±2048°/sec

Pinout



GP 6	IMU_SDA	I2C SDA	
GP 7	IMU_SCL	I2C SCL	
GP 8	LCD_DC	LCD Command/Data Selection	
GP 9	LCD_CS	LCD Chip Selection	
GP 10	LCD_CLK	LCD CLK	
GP 11	LCD_DIN	LCD MOSI	
GP 12	LCD_RST	LCD Reset	
GP 23	IMU_INT1	QMI8658C INT1	
GP 24	IMU_INT2	QMI8658C INT2	
GP 25	LCD_BL	LCD Backlight Control	
GP 29	BAT_ADC	Battery Voltage Acquisition Pin	biased by resistors to 1/2 value of the battery voltage

Dimensions

Arduino IDE Example

Environment Building

- Open Arduino IDE, and enter File -> Preferences.
- In the pop-up dialog box, enter the following URL in the "Additional Boards Manager URLs" field:

```
https://github.com/earlephilhower/arduino-pico/releases/download/global/package_rp2040_index.json
```

- Click "OK" to close the dialog box.
- Go to Tools -> Dev Board Manager in IDE.
- Type "pico" in the search box and select "Add".

Resource

Demo

- Demo code
- Clock project shared by Wienzek Daniel

Drawing

- Schematic diagram
- 3D drawing

Datasheet

- GC9A01A
- QMI8658C

Official Raspberry Pi Documents

- Raspberry Pi Pico MicroPython Book
- Raspberry Pi related books
- Pico datasheet
- RP1-PICO-R3-PUBLIC-SCHEMATIC
- Pico R3 A4 Pinout
- Getting started with pico
- Pico c sdk
- Pico python sdk.pdf
- Pico datasheet
- Rp2040 datasheet
- Hardware design with rp2040

Raspberry Pi Demo

- Raspberry Pi C/C++ (Github)
- Raspberry Pi Micropython (Github)

Developing Software

- Thonny

Download Firmware

- MicroPython Firmware Download

- C_Blink Firmware Download

Support

If you require technical support, please go to the page and open a ticket.